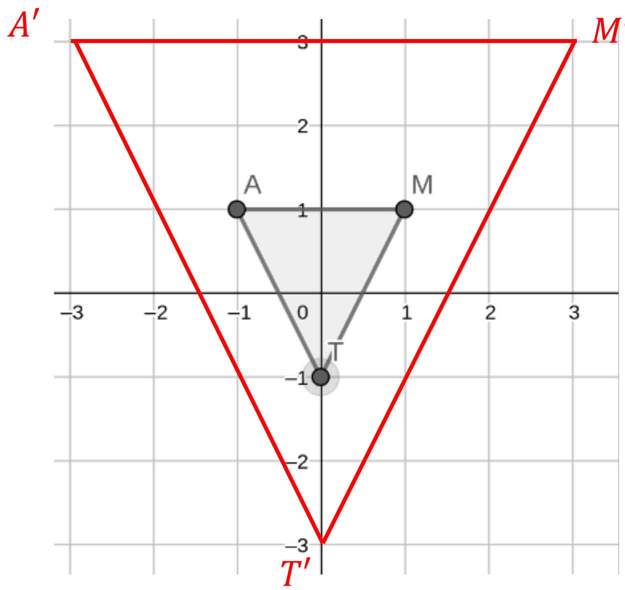
Grade 8 | Name_Answer Key________
Unit 9 | Date___________________
Lesson 8 Practice | Period__________________

$\triangle MAT$ is plotted on the coordinate grid. Use the grid for questions 1-4.



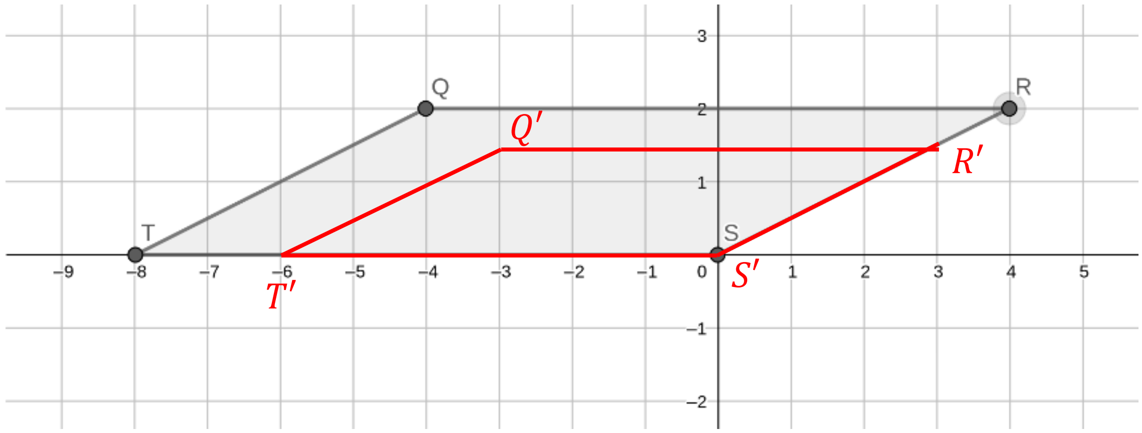
If the figure were dilated by a scale factor of 3 with the origin as the center of dilation, what are the coordinates for the image? Complete the table below with the coordinates of each vertex.

Coordinates		**Coordinates**	
1. Point M	$(1,1)$	**Point M'**	$(3,3)$
2. Point A	$(-1,1)$	**Point A'**	$(-3,3)$
3. Point T	$(0,-1)$	**Point T'**	$(0,-3)$

4. Plot $\triangle M'A'T'$ on the coordinate grid above.

5. Describe the relationship between the coordinates of the preimage and the image.
The coordinates of the image are 3 times the coordinates of the corresponding points on the preimage.

Quadrilateral $QRST$ is shown on the coordinate grid.



If the figure were dilated by a scale factor of $\frac{3}{4}$ with the origin as the center of dilation, what are the coordinates for the image? Complete the table below with the coordinates of each vertex.

Coordinates		**Coordinates**	
6. Point Q	$(-4,2)$	**Point Q'**	$\left(-3, 1\frac{1}{2}\right)$
7. Point R	$(4,2)$	**Point R'**	$\left(3, 1\frac{1}{2}\right)$
8. Point S	$(0,0)$	**Point S'**	$(0,0)$
9. Point T	$(-8,0)$	**Point T'**	$(-6,0)$

10. Describe the relationship between the coordinates of the preimage and the image.
The coordinates of the image are $\frac{3}{4}$ times the coordinates of the corresponding points on the preimage.